

Geometric Interaction Graph Neural Network for Predicting Protein–Ligand Binding Affinities from 3D Structures (GIGN)

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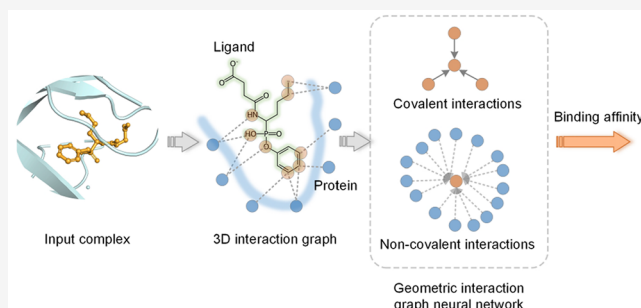


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ABSTRACT: Predicting protein–ligand binding affinities (PLAs) is a core problem in drug discovery. Recent advances have shown great potential in applying machine learning (ML) for PLA prediction. However, most of them omit the 3D structures of complexes and physical interactions between proteins and ligands, which are considered essential to understanding the binding mechanism. This paper proposes a geometric interaction graph neural network (GIGN) that incorporates 3D structures and physical interactions for predicting protein–ligand binding affinities. Specifically, we design a heterogeneous interaction layer that unifies covalent and noncovalent interactions into the message passing phase to learn node representations more effectively. The heterogeneous interaction layer also follows fundamental biological laws, including invariance to translations and rotations of the complexes, thus avoiding expensive data augmentation strategies. GIGN achieves state-of-the-art performance on three external test sets. Moreover, by visualizing learned representations of protein–ligand complexes, we show that the predictions of GIGN are biologically meaningful.



Drug discovery is an expensive process. Typically, a single drug can cost up to around 1 billion dollars and takes 10 years of development and testing before potentially being FDA approved.¹ Moreover, this process can fail at any point due to undesired side effects or experimental disproof of the promising therapeutic efficacy. A key problem in drug discovery is the prediction of the protein–ligand binding affinity (PLA), which describes the strength of the binding interaction between a ligand (drug candidate) and a protein (target). To identify effective and safe drugs for a given protein, pharmacologists have to test thousands of chemical compounds.² However, experimentally measuring the PLA is both time- and resource-consuming. In silico methods are able to accelerate the process of drug screening by ranking drug candidates and prioritizing the better ones according to the predictions of the PLA.

With the increasing amount of high-quality experimentally determined protein–ligand structures and their binding affinities, machine learning (ML) methods have been widely used to predict protein–ligand binding affinities or interactions by identifying useful patterns from data.^{2–7,7–27} Existing ML-based methods for PLA prediction can be divided into two categories, interaction-free and interaction-based methods, according to whether the models make decisions relying on physical interactions, as shown in Figure 1.

Interaction-free methods^{2,17–19,23–25,28–32} implicitly assume that ML models can predict PLA from data that do not reveal physical protein–ligand interactions. Therefore, ligands are often expressed by the simplified molecular-input line-entry

system (SMILES), or 2D graphs, and proteins are represented by sequences, while atomic interactions are omitted for simplicity. For example, DeepDTA¹⁷ uses SMILES strings and protein sequences as input from which two convolutional neural networks (CNNs) are used to extract features. On the other hand, GraphDTA¹⁸ and MGraphDTA² represent ligands as 2D molecule graphs to retain 2D structural information. The interaction-free methods can be used to predict binding affinities from complexes without 3D structure and physical interaction information. However, 3D structure and physical interaction priors have been demonstrated to be essential for improving the model generalization capability.³³

In contrast, interaction-based models^{3–7,7–16,34} make predictions based on the 3D structures of complexes and physical interactions of proteins and ligands. Among the interaction-based models, 3D-convolutional neural networks (3D-CNNs)^{5,12} and interaction graph neural networks (IGNNs)^{3,8,11,15,33} are most commonly used to predict binding affinities from 3D structures with atomic interaction information. 3D-CNNs express a complex as a 3D grid and extract the

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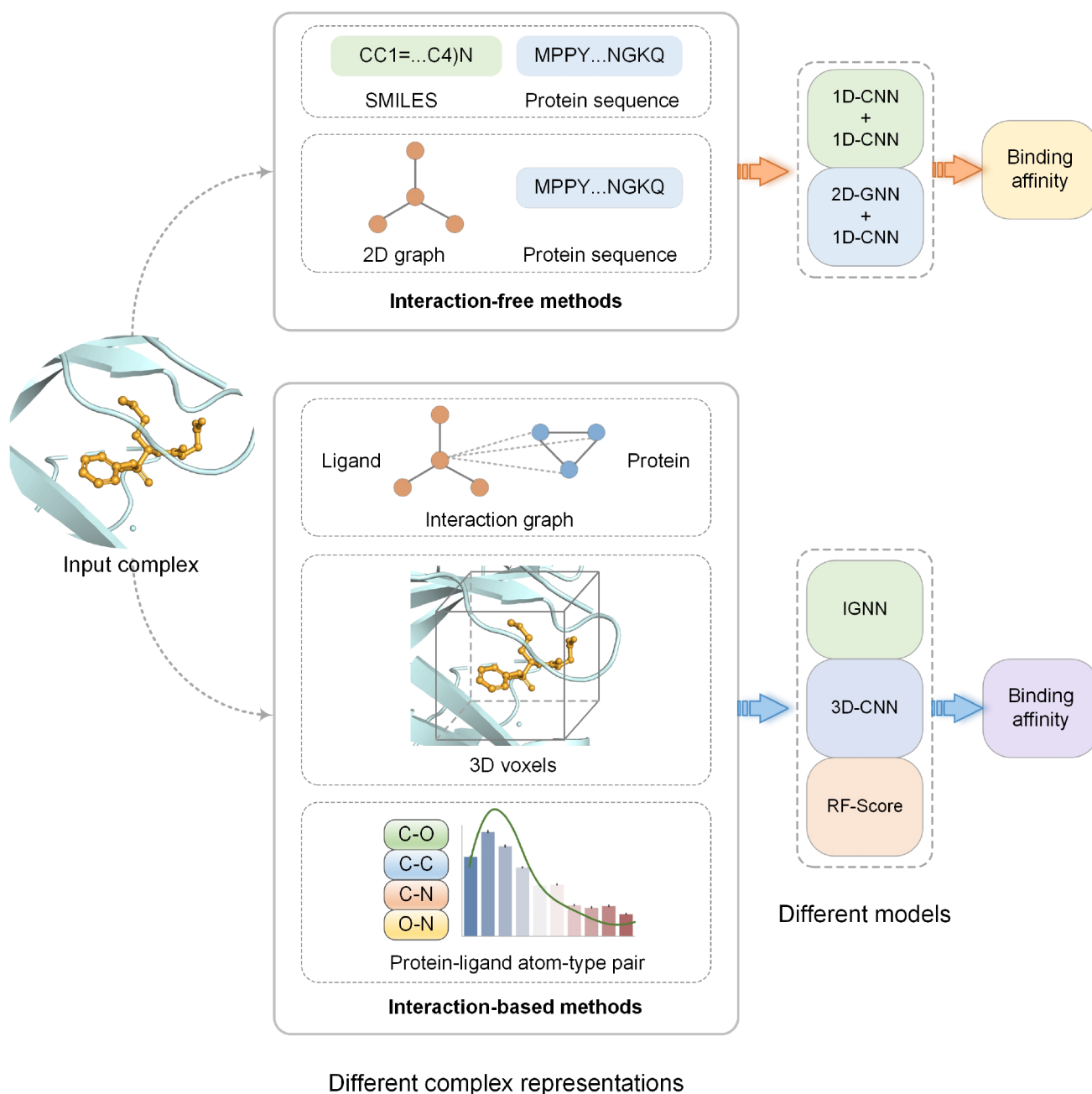


Figure 1. Interaction-free and interaction-based methods for PLA prediction.

structure and interaction information from the 3D grid using 3D convolutions, which is less efficient because most of the voxels do not contain useful information about the structure. Furthermore, since the position of each voxel is not rotationally invariant, rotating the atomic coordinates will change the prediction value of the binding affinity, which is inconsistent with biological fact. On the other hand, IGNNs represent protein–ligand complexes as interaction graphs, where nodes correspond to atoms and edges correspond to covalent or noncovalent bonds/interactions. The most common way to inject structure information into IGNNs is to use pairwise atom–atom distances computed from their 3D coordinates,^{3,11} which ensures IGNNs' invariance to translations and rotations of the complexes.

Despite the great potential of IGNNs in PLA prediction, they may not be ready for real applications due to their insufficient generalization ability.^{3,35} We summarized two potential problems of existing IGNNs in binding affinity prediction. First, existing IGNNs often regard covalent and noncovalent interactions as the same type of interaction.^{33,35–37} In this context, a ligand node can receive messages from its covalent and noncovalent neighbors simultaneously during message passing, as shown in Figure 2(a). Although this assumption simplifies the modeling, it has a significant disadvantage. The number of noncovalent interactions is greatly larger than that of covalent ones (see Supplement C for more details) and thus noncovalent interactions will dominate the computation (i.e., the information on covalent interactions may be engulfed by that on noncovalent ones). Second, the geometric prior (or

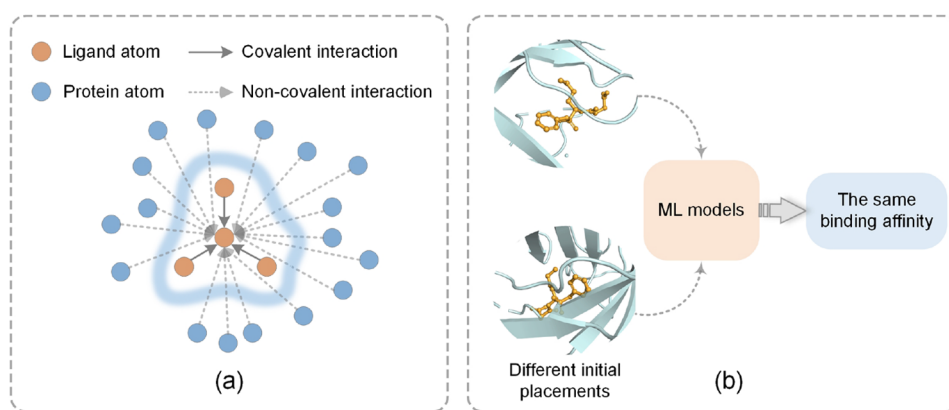


Figure 2. Motivation of the proposed method. (a) A ligand node receives messages from its covalent and noncovalent neighbors. In this example, the ligand node has 3 covalent neighbors and 17 noncovalent neighbors. Therefore, noncovalent interactions will dominate the computation. (b) The predicted binding affinity should be invariant to different initial 3D placements.

symmetry prior) is an important inductive bias that can be used to improve the model generalization ability by restricting neural networks to relevant functions through exploiting the symmetry of problems.^{1,38–40} For example, the predicted binding affinity should remain unchanged for the same protein–ligand complex with different initial placements (e.g., different positions and orientations), as shown in Figure 2(b). However, this invariance property still remains under-explored (e.g., there are fewer studies trying to demonstrate that IGNNs can benefit from considering invariance property).

To tackle the aforementioned issues, we propose a geometric interaction graph neural network (GIGN) that incorporates 3D structures and physical interactions with the invariance constraint to predict protein–ligand binding affinities. Key contributions of GIGN include the following:

- GIGN uses a heterogeneous interaction layer that unifies covalent and noncovalent interactions into the message passing phase to learn node representations more effectively. The heterogeneous interaction layer treats covalent and noncovalent interactions as different types of interactions and processes them independently during message passing, thus avoiding the drawback described in Figure 2(a).
- GIGN enforces the neural network to satisfy the invariance property with respect to input translations and rotations. We show that the model generalization ability can be greatly improved by considering the invariance property.
- The experimental results indicate that GIGN achieves state-of-the-art performance on three external test sets with a low computational cost, which is more readily applicable to large-scale databases.
- The visualization results show that GIGN can capture essential features relevant to binding.

PROBLEM DEFINITION

In this work, a complex is represented as a 3D interaction graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R}) = (\mathcal{V}_l \cup \mathcal{V}_p, \mathcal{E}_l \cup \mathcal{E}_p \cup \mathcal{E}_{lp}, \mathcal{R}_l \cup \mathcal{R}_p)$, where $\mathcal{V}_l$ and $\mathcal{V}_p$ are sets of ligand and protein nodes (i.e., atoms), respectively; $\mathcal{E}_l$ and $\mathcal{E}_p$ are sets of ligand and protein covalent edges (i.e., covalent bonds/interactions), respectively; $\mathcal{E}_{lp}$ is the set of noncovalent edges (i.e., noncovalent bonds/interactions); and $\mathcal{R}_l$ and $\mathcal{R}_p$ are sets of 3D coordinates of ligand

and protein nodes, respectively. Each node $v_i \in \mathcal{V}$ has its initial feature $x_i \in \mathbb{R}^n$ and 3D coordinate $r_i \in \mathbb{R}^3$. A covalent edge $e_{ij} \in \mathcal{E}_l \cup \mathcal{E}_p$ exists when atoms v_i and v_j are connected by covalent bonds. A noncovalent edge $e_{ik} \in \mathcal{E}_{lp}$ exists when $d_{ik} < 5$ Å, where $d_{ik} = \|r_i - r_k\|$ is the distance between the ligand atom $v_i \in \mathcal{V}_l$ and the protein atom $v_k \in \mathcal{V}_p$. The initial features used for nodes can be found in Supplement A Table S1. Given a 3D interaction graph, our goal is to learn a model f that satisfies invariance to predict the binding affinity as $f(\mathcal{G}) = \hat{y}$, where $\hat{y} \in \mathbb{R}^1$.

GRAPH NEURAL NETWORKS

Graph neural networks (GNNs) are permutation-equivariant networks that operate on graph data.^{39,41} Specifically, a GNN model uses a message passing function⁴¹ to iteratively update the representation of a node by aggregating messages from its neighboring nodes according to

$$m_{ji} = M(h_j^{(t)}, h_i^{(t)}) \quad (1)$$

$$m_i = \sum_{v_j \in \mathcal{N}(v_i)} m_{ji} \quad (2)$$

$$h_i^{(t+1)} = U(h_i^{(t)}, m_i) \quad (3)$$

where $h_i^{(t)}$ denotes the node representation of v_i at t -th layer of GNNs, $h_i^{(0)} = Wx_i$ ($W \in \mathbb{R}^{m \times n}$), $\mathcal{N}(v_i)$ is the set of neighbors of v_i , and M and U are differentiable functions such as multilayer perceptrons (MLPs).

INVARIANCE

Let $T: \mathcal{X} \rightarrow \mathcal{X}$ be a set of transformations (e.g., rotation, translation, reflection, or permutation). A graph neural network $f: \mathcal{X} \rightarrow \mathcal{Y}$ is invariant to T on $\mathcal{X}$ if

$$f(T(\mathcal{X})) = f(\mathcal{X}) \quad (4)$$

In this work we explore the following two types of invariance:

- Translation invariance, where translating the input by a translation vector g results in an invariant output (i.e., $f(X + g) = f(X)$).
- Rotation invariance, where rotating the input by a rotation matrix Q results in an invariant output (i.e., $f(QX) = f(X)$).

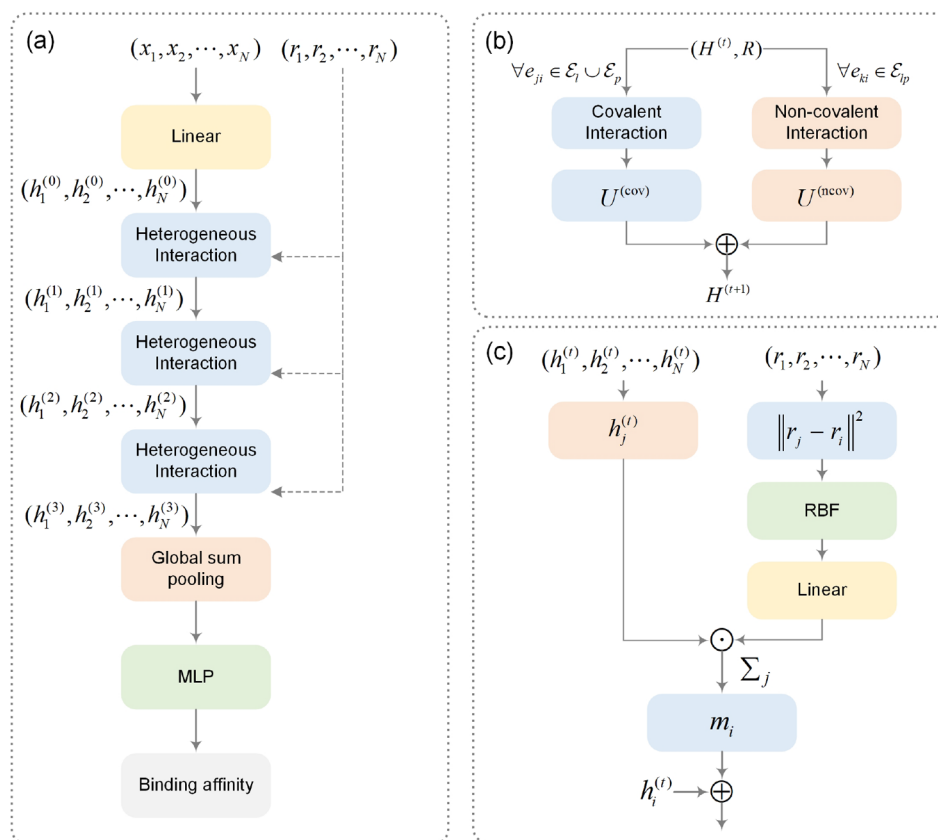


Figure 3. (a) The computational pipeline of GIGN. (b) The heterogeneous interaction layer. (c) The details of covalent and noncovalent interactions of the heterogeneous interaction layer.

GEOMETRIC INTERACTION GRAPH NEURAL NETWORK

The overall computational pipeline of GIGN is depicted in Figure 3(a). GIGN uses a heterogeneous interaction layer that unifies covalent and noncovalent interactions into the message passing phase to learn node representations more effectively, as shown in Figure 3(b). Specifically, for a 3D interaction graph $\mathcal{G}$ with node features $(x_1, x_2, \dots, x_N)$ and 3D coordinates $(r_1, r_2, \dots, r_N)$, the heterogeneous interaction layer can be defined as

$$m_{ji}^{(\text{cov})} = M^{(\text{cov})}(h_j^{(t)}, h_i^{(t)}, \lambda(\|r_j - r_i\|^2)), \forall e_{ji} \in \mathcal{E}_1 \cup \mathcal{E}_p \quad (5)$$

$$m_{ki}^{(\text{ncov})} = M^{(\text{ncov})}(h_k^{(t)}, h_i^{(t)}, \lambda(\|r_k - r_i\|^2)), \forall e_{ki} \in \mathcal{E}_{\text{lp}} \quad (6)$$

$$m_i^{(\text{cov})} = \sum_{v_j \in \mathcal{N}(v_i)} m_{ji}^{(\text{cov})}, \forall e_{ji} \in \mathcal{E}_1 \cup \mathcal{E}_p \quad (7)$$

$$m_i^{(\text{ncov})} = \sum_{v_k \in \mathcal{N}(v_i)} m_{ki}^{(\text{ncov})}, \forall e_{ki} \in \mathcal{E}_{\text{lp}} \quad (8)$$

$$h_i^{(t+1)} = U^{(\text{cov})}(h_i^{(t)}, m_i^{(\text{cov})}) + U^{(\text{ncov})}(h_i^{(t)}, m_i^{(\text{ncov})}) \quad (9)$$

where $M^{(\text{cov})}$ and $M^{(\text{ncov})}$ are implemented as $M^{(\text{cov})}(h_j^{(t)}, h_i^{(t)}, \lambda(\|r_j - r_i\|^2)) = h_j^{(t)} \odot [\lambda(\|r_j - r_i\|^2)]$ and $M^{(\text{ncov})}(h_k^{(t)}, h_i^{(t)}, \lambda(\|r_k - r_i\|^2)) = h_k^{(t)} \odot [\lambda(\|r_k - r_i\|^2)]$ ($\odot$ is the element-wise product), respectively, as shown in Figure 3(c), $U^{(\text{cov})}$ and $U^{(\text{ncov})}$ are implemented as MLPs, and λ is the

linear combination of Gaussian radial basis functions (RBFs)⁴² $\exp\left(-\frac{1}{\sigma^2} \|d_{ji} - \mu_k\|^2\right)$ located at centers $0 \text{ \AA} \leq \mu_k \leq 6 \text{ \AA}$ every $2/3 \text{ \AA}$ with $\sigma = 2/3 \text{ \AA}$. This is chosen to guarantee all distances within $5 \text{ \AA}$ are covered. Through eqs 5 and (6), we inject the structure information $\lambda(\|r_j - r_i\|^2)$ and $\lambda(\|r_k - r_i\|^2)$ into the model. Intuitively, $m_i^{(\text{cov})}$ and $m_i^{(\text{ncov})}$ are at different scales since the number of noncovalent interactions is greatly larger than that of covalent interactions. Therefore, we use $U^{(\text{cov})}$ and $U^{(\text{ncov})}$ to convert $m_i^{(\text{cov})}$ and $m_i^{(\text{ncov})}$ into the same scale before integrating them, as shown in eq 9, which avoids the issue described by Figure 2 (a). Note that the noncovalent interactions within the ligand (or the protein) are also included because the receptive field of GNNs can be expanded with increasing layers^{2,43} (i.e., an atom can access its k -hop neighbors for a k -layer GNN).

ANALYSIS ON INVARIANCE FOR GIGN

Note that the heterogeneous interaction layer F is invariant to translations and rotations. Formally, F satisfies

$$H^{(t+1)} = F(R, H^{(t)}) = F(QR + g, H^{(t)}) \quad (10)$$

where $H = (h_1, \dots, h_N) \in \mathbb{R}^{N \times m}$ is the node representation matrix of a graph with N nodes, $R = (r_1, \dots, r_N) \in \mathbb{R}^{N \times 3}$ is the 3D coordinate matrix, $g \in \mathbb{R}^3$ is a translation vector, and $Q \in \mathbb{R}^{3 \times 3}$ is a rotation matrix (orthogonal matrix).

The initial node feature matrix $X = (x_1, \dots, x_N) \in \mathbb{R}^{N \times n}$ is invariant to transformations (translations and rotations) on R ; in other words, we do not inject any position information about R

into X . Therefore, $H^{(0)} = WX$ ($W \in \mathbb{R}^{m \times n}$) is also invariant to transformations. Given that distance between two nodes is invariant to translations $\|(r_j + g) - (r_i + g)\|^2 = \|r_j - r_i\|^2$ and to rotations $\|Qr_j - Qr_i\|^2 = [Q(r_j - r_i)]^T [Q(r_j - r_i)]$, we can

$$\begin{aligned} &= (r_j - r_i)^T Q^T Q (r_j - r_i) \\ &= \|r_j - r_i\|^2 \end{aligned}$$

conclude that the message $m_{ji}^{(\text{cov})}$ in eq 5 is also invariant:

$$\begin{aligned} m_{ji}^{(\text{cov})} &= M^{(\text{cov})}(h_j^{(t)}, h_i^{(t)}, \lambda(\|(Qr_j + g) - (Qr_i + g)\|^2)) \\ &= M^{(\text{cov})}(h_j^{(t)}, h_i^{(t)}, \lambda(\|r_j - r_i\|^2)) \end{aligned} \quad (11)$$

Similarly, $m_{ji}^{(\text{ncov})}$ in eq 6 is invariant. Therefore, $m_i^{(\text{cov})}$, $m_i^{(\text{ncov})}$, and $h_i^{(t+1)}$ in eqs 7–9 are also invariant because they only depend on $m_{ji}^{(\text{cov})}$, $m_{ji}^{(\text{ncov})}$, and $h_i^{(t)}$, respectively. Thus, we can conclude that $F(QR + g, H^{(t)}) = F(R, H^{(t)})$.

BINDING AFFINITY PREDICTION

We use a global sum pool to encapsulate the node-level representations at the last graph convolution layer (T -th layer) into a graph-level vector as

$$h_G = \sum_{v_i \in V} h_i^{(T)} \quad (12)$$

An MLP ($\mathbb{R}^m \rightarrow \mathbb{R}^1$) is used to map h_G to a scalar $\hat{y} \in \mathbb{R}^1$ indicating the binding affinity. The model is trained to minimize a mean-squared error loss function as follows:

$$L_{\text{mse}} = \frac{1}{|\mathcal{D}|} \sum_{n=1}^{|\mathcal{D}|} (\hat{y}_n - y_n)^2 \quad (13)$$

where y_n is the experimental binding affinity of n -th sample in data set $\mathcal{D}$ and $|\mathcal{D}|$ is the sample size.

IMPLEMENTATION DETAIL

Experiments are conducted using an NVIDIA GeForce RTX A4000 with 16 GB memory. An Adam optimizer with a 5×10^{-4} learning rate is used to update model parameters. We apply a weight decay of 1×10^{-6} for GIGN to reduce overfitting. The batch size is set to 128. The source code for GIGN is available at <https://github.com/guaguabujianle/GIGN>.

DATA SET PREPARATION

The proposed GIGN is trained and validated on PDBbind ver. 2016. 13 285 complexes (general and refined sets) with experimental binding affinities expressed as pK_a ($-\log K_d$ or $-\log K_i$) values are collected from the PDBbind website (<http://www.pdbbind.org.cn/>). We remove 381 samples that can not be processed by RDkit. The remaining 12904 samples are randomly split into training ($N = 11904$) and validation ($N = 1000$) sets. Three independent external test sets, the PDBbind 2013 core set ($N = 107$), the 2016 core set ($N = 285$), and the 2019 holdout set ($N = 4366$), are used to test the generalization capability of GIGN. The 2013 and 2016 core sets are two commonly used benchmarks to evaluate the performance of binding affinity prediction.^{3,5,10,14,35} However, their small sample sizes tend to result in overly optimistic results.⁴ Therefore, we collect 4366 samples from PDBbind ver. 2019

that are unavailable in the other four sets as a new external holdout set, mimicking a real temporal split scenario in which binding affinities for newly released structures are predicted by a model trained on past structural data. This temporal split setting is a more difficult task and also more precisely reflects the real situation of drug discovery.⁴⁴ Note that there is no overlap of the complexes between the five sets. To demonstrate that GIGN can be generalized to the non-PDBbind data set, we also validate it in the CSAR-HiQ data set.⁴⁵ The CSAR-HiQ data set contains two subsets with 343 complexes in total. After removing the complexes that overlap with the above five sets and can not be processed by RDkit, we retain 47 samples as a non-PDBbind external test set. During the processing, ligand–protein atom pairs with a distance greater than 5 Å are removed, following previous studies.^{11,33}

PERFORMANCE INDICATORS

The model performance is evaluated by Pearson's correlation coefficient (R_p) and the root-mean-square error (RMSE), which are defined as follows.

$$\text{RMSE} = \sqrt{\frac{1}{|\mathcal{D}|} \sum_{n=1}^{|\mathcal{D}|} (\hat{y}_n - y_n)^2} \quad (14)$$

$$R_p = \frac{\sum_{i=1}^{|\mathcal{D}|} (\hat{y}_n - \bar{\hat{y}})(y_n - \bar{y})}{\sqrt{\sum_{i=1}^{|\mathcal{D}|} (\hat{y}_n - \bar{\hat{y}})^2 (y_n - \bar{y})^2}} \quad (15)$$

BASELINES

We compare GIGN with the following representative methods.

- Interaction-free methods:
 - DeepDTA¹⁷ represents ligands and proteins as SMILES and sequences, respectively, and establishes two CNNs to extract features from SMILES and sequences, respectively.
 - GraphDTA¹⁸ is similar to DeepDTA but represents ligands as 2D graphs and uses GNNs (including GCN, GAT, GIN, and GAT-GCN) to learn representations from graphs.
 - MGraphDTA² also represents ligands as 2D graphs and uses a multiscale GNN to learn representations from graphs.
- Interaction-based methods:
 - Pafnucy⁵ is a representative 3D-CNN model that can learn the spatial structure and physical interactions of protein–ligand complexes.
 - OnionNet¹⁴ generates interaction-based features based on contact numbers in protein–ligand interaction shells and uses 2D-CNN to learn representations for prediction.
 - PotentialNet¹⁵ presents a staged gated graph neural network to include both intramolecular interactions and noncovalent interactions between ligands and proteins.
 - GNN-DTI¹¹ is a GNN-based method that incorporates the 3D structural information and uses distance-aware graph attention to differentiate the contribution of each interaction to the binding affinity.

Table 1. Comparison Results of the Proposed GIGN and Baselines on Three External Test Sets

model		2013 core set ($N = 107$)		2016 core set ($N = 285$)		2019 holdout set ($N = 4366$)	
		RMSE	R_p	RMSE	R_p	RMSE	R_p
interaction-free methods	DeepDTA	1.639 (0.026)	0.718 (0.014)	1.357 (0.015)	0.785 (0.007)	1.485 (0.023)	0.586 (0.012)
	GCN	1.749 (0.062)	0.662 (0.032)	1.513 (0.048)	0.719 (0.023)	1.763 (0.039)	0.439 (0.021)
	GAT	2.043 (0.029)	0.476 (0.022)	1.748 (0.019)	0.594 (0.010)	1.663 (0.027)	0.432 (0.016)
	GIN	1.691 (0.124)	0.694 (0.059)	1.470 (0.065)	0.743 (0.027)	1.676 (0.032)	0.472 (0.021)
	GAT-GCN	1.645 (0.085)	0.711 (0.036)	1.434 (0.064)	0.754 (0.025)	1.705 (0.075)	0.474 (0.028)
interaction-based methods	MGraphDTA	1.680 (0.093)	0.696 (0.046)	1.439 (0.047)	0.753 (0.022)	1.553 (0.028)	0.538 (0.013)
	Pafnucy	1.517 (0.014)	0.783 (0.005)	1.450 (0.047)	0.769 (0.019)	1.438 (0.016)	0.612 (0.014)
	OnionNet	1.583 (0.079)	0.741 (0.037)	1.399 (0.076)	0.770 (0.027)	1.510 (0.034)	0.573 (0.014)
	PotentialNet	1.607 (0.027)	0.773 (0.010)	1.503 (0.033)	0.772 (0.007)	1.514 (0.028)	0.564 (0.014)
	GNN-DTI	1.533 (0.084)	0.767 (0.040)	1.384 (0.013)	0.779 (0.008)	1.446 (0.006)	0.614 (0.007)
	IGN	1.428 (0.020)	0.807 (0.001)	1.269 (0.030)	0.821 (0.013)	1.410 (0.015)	0.630 (0.008)
	SchNet	1.570 (0.029)	0.754 (0.030)	1.390 (0.023)	0.787 (0.016)	1.522 (0.071)	0.560 (0.028)
	EGNN	1.498 (0.025)	0.782 (0.015)	1.289 (0.021)	0.816 (0.011)	1.399 (0.013)	0.628 (0.010)
	GIGN (ours)	1.380 (0.009)	0.821 (0.003)	1.190 (0.017)	0.840 (0.007)	1.393 (0.007)	0.641 (0.006)

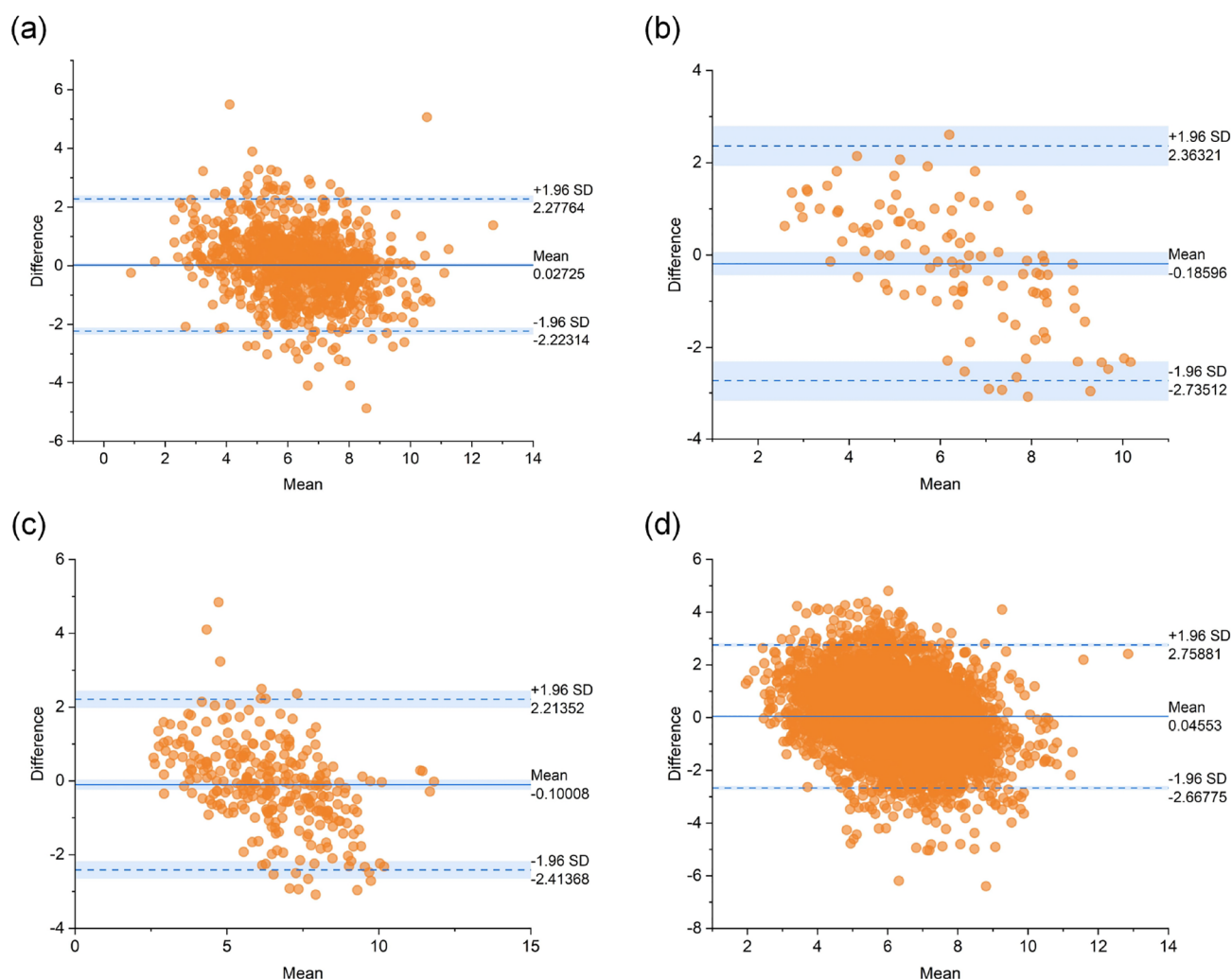


Figure 4. Bland–Altman plots for comparing predicted and experimental binding affinities for GIGN on the (a) validation set, (b) 2013 core set, (c) 2016 core set, and (d) 2019 holdout set, where the x-axis represents the mean (i.e., $(\hat{y}_n + y_n)/2$) and the y-axis represents the difference (i.e., $\hat{y}_n - y_n$).

- IGN³ designs two stacked GNN modules to sequentially learn the intramolecular and inter-molecular interactions, respectively.

- SchNet⁴² uses continuous filter convolutional layers to model local correlations, and it is invariant to translations and rotations.

Table 2. Experimental Results of the Ablation Study on Three External Test Sets

model	2013 core set		2016 core set		2019 holdout set	
	RMSE	R_p	RMSE	R_p	RMSE	R_p
GIGN_TMP	1.460 (0.008)	0.791 (0.003)	1.242 (0.015)	0.822 (0.004)	1.409 (0.004)	0.634 (0.004)
GIGN_TRVar	1.518 (0.047)	0.763 (0.019)	1.416 (0.024)	0.760 (0.010)	1.530 (0.004)	0.550 (0.007)
GIGN_RVar	1.520 (0.010)	0.765 (0.007)	1.344 (0.033)	0.788 (0.013)	1.447 (0.023)	0.608 (0.006)
GIGN (ours)	1.380 (0.009)	0.821 (0.003)	1.190 (0.017)	0.840 (0.007)	1.393 (0.007)	0.641 (0.006)

- EGNN³⁹ achieves state-of-the-art performance by exploring 3D translation and rotation symmetries.

All the baselines are implemented using the source code provided by the original papers. Note that GNN-DTI, SchNet, and EGNN were originally not designed for PLA predictions. Therefore, we tailor them to fit in with the PLA prediction task.

OVERALL COMPARISON

All the models share the same training, validation, and external test sets. All experiments are repeated three times, each with a different random seed. The validation set is used to select the best model with the lowest validation RMSE. The selected best model is used to evaluate the PLA prediction performance on three external test sets. We finally report the mean and standard deviation of performances based on three independent runs.

As shown in Table 1, it is observed that GIGN achieves the best performance on three external data sets, with 1.7%, 2.3%, and 1.7% improvements over the best baseline model in terms of R_p on the validation set, the 2013 core set, the 2016 core set, and the 2019 holdout set, respectively. Figure 4 shows the Bland–Altman plots for GIGN in the validation set, the 2013 core set, the 2016 core set, and the 2019 holdout set, where the differences between the predicted and experimental binding affinities are plotted against the averages of the affinities. The mean difference is the estimated bias, and the standard deviation (SD) of the differences measures the random fluctuations around this mean. It is observed that the mean differences for the four data sets are all around zero, which means that the predicted binding affinity is generally less biased. There are 94.60%, 95.37%, 95.08%, and 94.71% points within 95% limits of agreement (mean difference ± 1.96 standard deviation of the difference) for the four sets, which indicates good agreement between the predicted and experimental binding affinities. Moreover, we have the following observations.

As shown in Table 1, among the interaction-free methods, GraphDTA and MGraphDTA show worse performance than DeepDTA. However, these two 2D graph-based methods have shown superior performance compared to SMILES-based methods (e.g., DeepDTA) in Davis ($N = 30056$) and KIBA ($N = 118254$) data sets,^{2,17} which are much larger than PDBbind ver. 2016 ($N = 13285$) in terms of sample size. This result suggests that 2D graph-based methods easily overfit the training data when the sample size is relatively small. On the other hand, we can observe that models that incorporate 3D structure and physical interactions (i.e., PotentialNet, GNN-DTI, IGN, and GIGN) are superior to those 2D graph-based methods. This result suggests that, when data are scarce, it is necessary to inject the structure and interaction inductive biases into ML models to avoid learning these priors from insufficient amounts of data.

It is worth noting that all the methods only show a moderate generalization ability on the more difficult PDBbind ver. 2019 holdout set. This result suggests that accurately predicting PLA is still a challenge. Given that a threshold value of 0.600 for R_p is

typically used to qualify a good predictive quantitative structure–activity relationship model,^{35,46} only Pafnucy, GNN-DTI, IGN, EGNN, and GIGN can be considered satisfactory.

To explore why models achieve worse performance in the 2019 holdout set, we calculate the protein overlap rate and the ligand overlap rate between the test set and the training set. The overlap rate indicates the percentage of proteins (or ligands) in the test set that can also be found in the training set. The protein overlap rates are 100.0%, 99.6%, and 68.9% for the 2013 test set, the 2016 test set, and the 2019 holdout set, respectively. The ligand overlap rates are 37.4%, 30.9%, and 24.8% for the three sets. Obviously, the protein and ligand overlap rates for the 2013 test set and the 2016 test set are much higher than those for the 2019 holdout set, which can explain why the models achieve inferior performance in the 2019 holdout set. Although the complexes in the 2013 and 2016 test sets are completely different from the training set, the high overlap rate of proteins and ligands may cause information leakage, thus leading to overoptimistic results.

ABLATION STUDY

To verify the effectiveness of the proposed heterogeneous interaction layer and the invariance constraint, we compare GIGN with its three variants on the three benchmarks.

- GIGN_TMP uses traditional message passing where covalent and noncovalent interactions are considered the same type of interaction.
- GIGN_VarTR is variant to translations and rotations. Specifically, we add the corresponding coordinate r_i ($\times 0.1$ to stabilize training) to the initial feature x_i for node v_i (i.e., $(r_i \times 0.1) \parallel x_i$, where $\parallel$ is concatenation) to inject structural information to the initial node features. Obviously, these coordinates are variant to translations and rotations and thus $(r_i \times 0.1) \parallel x_i$ is also variant to the transformations.
- GIGN_VarR is only variant to rotations. The coordinates of the complex are normalized by subtracting the centroid of the ligand before adding them to the initial feature x_i .

Table 2 summarizes the experimental results of the ablation study. It is observed that GIGN exceeds the three variants, demonstrating the necessity of the heterogeneous interaction setting and the invariance constraint. Specifically, GIGN_VarR surpasses GIGN_VarTR greatly in the 2016 core set and the 2019 holdout set, which proves the importance of translational invariance. Without translational invariance, the model is less able to discern the same protein–ligand complex in different positions (i.e., it predicts different binding affinities for the same complexes in different positions). GIGN_VarR solves this problem by normalizing the coordinates, leading to a better generalization capability. Moreover, it can be observed that GIGN outperforms GIGN_VarR by a large margin, indicating the significance of rotational invariance. Another way to mitigate

rotational variance is to use data augmentation (i.e., randomly rotating the input complex during training), which is the strategy that Pafnucy adopts. GIGN enforces the model to recognize the same protein–ligand complex with different initial placements via invariance constraint, thus avoiding expensive data augmentation strategies.

■ PERFORMANCE EVALUATION ON THE NON-PDBBIND EXTERNAL TEST SET

To further evaluate the generalization ability of the IGNNs, we also evaluate them on the CSAR-HiQ data set. As shown in Table 3, IGNN, SchNet, EGNN, and GIGN achieve promising

Table 3. Performance Evaluation for Six IGNNs on the CSAR-HiQ Dataset

model	RMSE	R_p
PotentialNet	1.761 (0.073)	0.647 (0.037)
GNN-DTI	1.740 (0.094)	0.657 (0.049)
IGN	1.618 (0.130)	0.714 (0.057)
SchNet	1.587 (0.077)	0.746 (0.016)
EGNN	1.538 (0.010)	0.760 (0.012)
GIGN (ours)	1.498 (0.058)	0.774 (0.025)

performance in this non-PDBbind external test set, which suggests that the high performance on the PDBbind data set may be translated into high performance in real-world virtual screening and drug development.

■ GIGN RETAINS ITS SUPERIORITY IN REDOCKED STRUCTURES

There is a risk that a model trained only with the experimental structures cannot be generalized to redocked structures¹⁴ (i.e., the robustness of the model may be questionable). To assess the robustness of GIGN, we also evaluate it using redocked structures. Specifically, for each complex in the 2016 test set, we select a redocked structure with the lowest RMSD (RMSD < 2 Å, also called native-like pose) from the CASF-2016 benchmark.⁴⁷ We use GIGN to predict the binding affinities of these redocked structures. The R_p and RMSE values for the predicted binding affinities of redocked structures against the experimental binding affinity are 0.826 (0.005) and 1.261 (0.014), respectively. These results suggest that the predicted binding affinities for redocked structures are also close to experimental binding affinities.

■ PERFORMANCE EVALUATION UNDER THE COLD START SETTING

To further prevent the structural information on proteins or ligands from leaking into the test set, we split the original

training set ($N = 11904$) into new training, validation, and test sets in a ratio of 8:1:1 according to 30% protein sequence identity or ligand scaffold (also called cold start setting²). We use the random split setting as a control group. It is not surprising that the models achieve much worse performance in the cold start setting, as shown in Table 4, because the cold start setting prevents the structural information on proteins or ligands from leaking into the test set. Moreover, it can be observed that GIGN achieves better performance in the random and scaffold split settings, while it obtains relatively inferior performance in the sequence identity split setting. This is because the heterogeneous interaction layer of GIGN is mainly designed to enhance the node representations of ligand atoms instead of protein atoms.

■ EPISTEMIC UNCERTAINTY QUANTIFICATION OF GIGN

ML-based methods may produce unreliable predictions for those out-of-domain complexes. One possible way to reduce unreliable predictions is to filter predictions according to their estimated uncertainties.⁸ We use Monte Carlo dropout (MCDO) to quantify epistemic uncertainties⁴⁸ of the predictions. In particular, we estimate the uncertainties of predictions for the 2019 holdout set, which has a relatively large sample size. The predictions are first ranked according to the estimated uncertainties. We then select one-third of the samples with the lowest uncertainties (low group) and one-third of the samples with the highest uncertainties (high group). As a comparison, we randomly select one-third of the samples from the 2019 holdout set and label it as a random group. A baseline group that simply uses all samples in the data set is also provided for a better illustration. Figure 5(a) shows the comparison between the four groups in terms of R_p . We can observe that the R_p of the low group is much larger than those of the random and low groups, which suggests the estimated uncertainties can be used to filter unreliable predictions.

To figure out why ML models make some predictions with high uncertainty, we first plot the histogram of experimental binding affinities for the training set and the scatter diagram between the experimental binding affinity and the estimated uncertainty for the 2019 holdout set, as shown in Figure 5(b) and (c). It is observed that the experimental binding affinities generally follow a normal distribution, as shown in Figure 5(b). Intuitively, the model would show high uncertainty for those complexes with low/high experimental binding affinities due to the smaller sample size. However, the result illustrated in Figure 5(c) reveals that smaller sample size is not the main reason for the high uncertainties of the predictions because the estimated uncertainties for those complexes with low/high experimental binding affinities are not much higher than others. A possible

Table 4. Performance evaluation for six IGNNs in the random split and cold start (split based on sequence identity and scaffold) scenarios

model	random		sequence identity (30%)		scaffold	
	RMSE	R_p	RMSE	R_p	RMSE	R_p
PotentialNet	1.360 (0.030)	0.691 (0.013)	1.582 (0.017)	0.507 (0.006)	1.538 (0.006)	0.532 (0.015)
GNN-DTI	1.403 (0.011)	0.666 (0.007)	1.499 (0.008)	0.562 (0.006)	1.401 (0.027)	0.631 (0.018)
IGN	1.295 (0.011)	0.732 (0.006)	1.509 (0.034)	0.561 (0.025)	1.361 (0.041)	0.665 (0.023)
SchNet	1.398 (0.044)	0.672 (0.021)	1.499 (0.022)	0.566 (0.020)	1.456 (0.046)	0.607 (0.014)
EGNN	1.285 (0.025)	0.730 (0.012)	1.483 (0.029)	0.579 (0.017)	1.395 (0.036)	0.635 (0.029)
GIGN (ours)	1.264 (0.015)	0.746 (0.006)	1.521 (0.012)	0.565 (0.007)	1.353 (0.002)	0.685 (0.004)

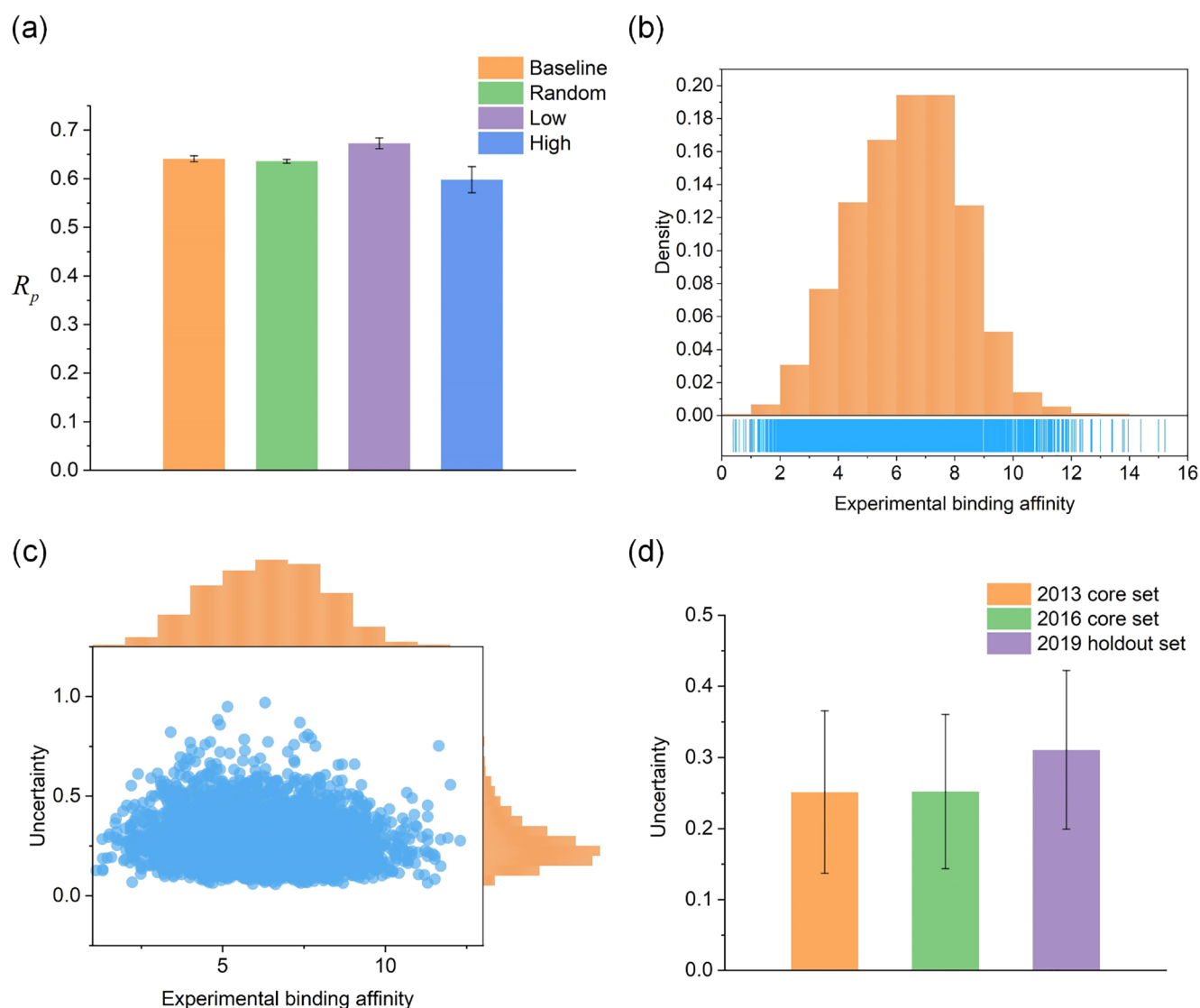


Figure 5. Uncertainty estimation of predictions for GIGN. (a) R_p values of baseline, random, low, and high groups on the 2019 holdout set (baseline, using the whole data set; random, randomly selecting one-third of the samples; low, the one-third of the samples with the lowest uncertainties; and high, the one-third of the samples with the highest uncertainties). (b) Histogram of experimental binding affinities for the training set. (c) Scatter diagram of uncertainty against experimental binding affinity for the 2019 holdout set. (d) Estimated uncertainties for predictions in the 2013 core set, the 2016 core set, and the 2019 holdout set.

reason for high uncertainties is that some inputs are out-of-domain complexes. Concretely, Figure 5(d) depicts the estimated uncertainties for predictions in the 2013 core set, the 2016 core set, and the 2019 holdout set (the distribution of the estimated uncertainty is shown in Supplement B Figure S1). It can be observed that the estimated uncertainties for the 2019 holdout set are much higher than those for the 2013 and 2016 core sets. The 2019 holdout set is collected following the temporal split setting, and it contains newly released structures that may exhibit different patterns (i.e., out-of-domain complexes) compared to complexes in the training set, which can explain why the estimated uncertainties for it are higher than those for the 2013 and 2016 core sets and why GIGN achieves relatively worse performance in this data set.

■ VISUAL EXPLANATIONS FOR GIGN

GNNs cannot be fully trusted without understanding and verifying their working mechanisms,^{43,49} which limits their

application in drug discovery scenarios. In this section, we conduct two visual explanation experiments to rationalize GIGN.

To demonstrate that the learned graph-level representations of complexes (i.e., h_G in eq 12) are meaningful, we visualize them at some stages during training. Specifically, we randomly select 256 complexes from PDBbind ver. 2016 and use GIGN to map them to m -dimensional vectors ($m = 256$ in our study). The 256-dimensional vectors, similar to the fingerprint representation of a complex, are reduced to 2D embeddings by t-SNE.⁵⁰ Figure 6 shows the distribution of 2D embeddings on epochs 1, 80, 175, and 302 (the best epoch with the lowest validation RMSE). As shown in Figure 6, the scatter plots show some degree of chaos at the beginning and then group into clusters during training according to the binding affinities, which suggests that the learned graph representations are meaningful.

To further explore which ligand atoms provide the most significant contribution to PLA prediction, we visualize the

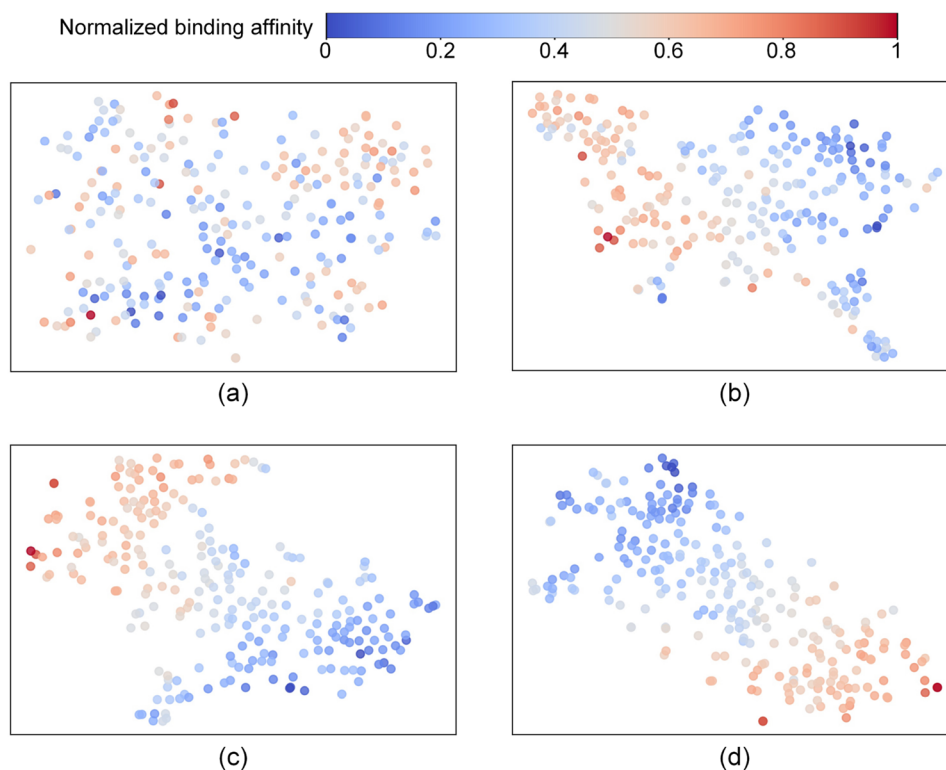


Figure 6. Visualizations of complex embeddings at epochs (a) 1, (b) 80, (c) 175, and (d) 302.

ligand atom importance using Grad-AAM.² Figure 7 shows the visualization results, where the distribution of atom importance shows a close match to the binding mode (more examples can be found in Supplement B Figure S2). Specifically, we have two following observations. First, the ligand atoms with higher weights tend to locate at positions with hydrogen bonds that are generally considered to be facilitators of protein–ligand binding. Second, those ligand atoms with lower weights are generally outside the binding pocket, which suggests that GIGN is sensitive to binding poses. We also provide a case about visualizing protein atom importance using Grad-AAM (Figure 8). Similarly, it can be observed that the atom importance has a relationship with hydrogen bonds. Overall, these results suggest that GIGN may be able to capture essential features relevant to binding.

■ PARAMETER SENSITIVITY ANALYSIS

In this experiment, two major hyper-parameters of GIGN are tested, which are the number of heterogeneous interaction layers T and the number of hidden channels m . We analyze the effect of T ranging from 1 to 6. As shown in Figure 9, the model achieves better performance when using 2–4 heterogeneous interaction layers. Theoretically, with the increase of T , the receptive field of GNN can be enlarged⁴³ (i.e., the model can detect larger molecular substructures). However, increasing T would also cause the oversmoothing problem.⁴³ Therefore, applying three heterogeneous layers for GIGN is already enough to capture sufficient structural information. We analyze the effect of m by setting it to different values (32, 64, 128, 192, 256, and 320). As shown in Figure 9, a general upward trend can be observed by increasing the value of m from 32 to 256. This is because the relatively larger hidden channels can encode more information. However, a slight degradation of performance can be found if we

continue to increase it to 320, which is probably caused by the overfitting problem.

■ MODEL COMPLEXITY

Besides the state-of-the-art performance on PLA prediction, another advantage of GIGN is the low computational cost. Figure 10 shows the number of parameters and floating point operations (FLOPs) for six IGNNs (PotentialNet, GNN-DTI, IGN, SchNet, EGNN, and GIGN). Among the top three IGNNs (IGN, EGNN, and GIGN) that have better performance than the others, as shown in Table 1, GIGN has a much lower computational cost than IGN and EGNN. This lightweight advantage implies that GIGN is more readily applicable to large-scale databases.

■ LIMITATIONS

One disadvantage of interaction-based models is that they heavily rely on optimized structures. Specifically, all complexes in the PDBbind data set are optimized according to the minimum energy criterion, which means that the ligand binding structure (i.e., the best position and orientation of the ligand) should be provided in order to train the interaction-based models. However, optimized structures may not always be available in real applications. Luckily, EQUIBIND¹ shows that it is possible to use deep learning to predict ligand binding structures with significant speed-ups and better quality than traditional docking. If the optimized structures are unknown, EQUIBIND can be a useful tool to predict how a ligand binds to the protein.

This work focuses on the study of how to effectively incorporate 3D structures of complexes and protein–ligand physical interactions into GNNs. To this end, we propose a simple but effective GNN-based model, GIGN, to predict the

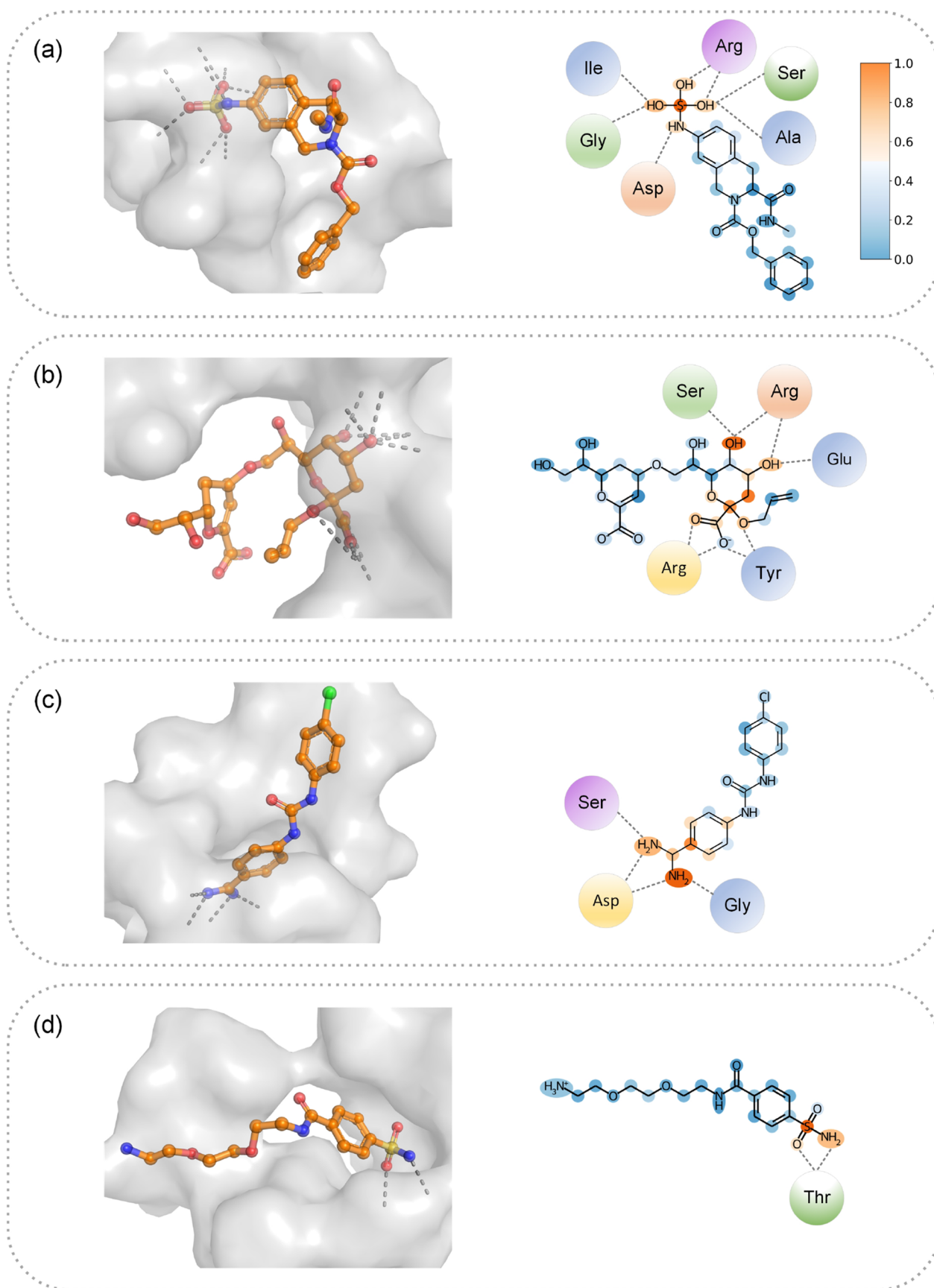


Figure 7. Visualizations of atom importance of ligands using Grad-AAM for complexes with PDB IDd (a) 2f6v, (b) 2r1x, (c) 1bjv, and (d) 1cnx, where the left and right columns are 3D and 2D binding modes with hydrogen bonds, respectively. The atom importance is shown with the blue/orange color in the 2D binding mode.

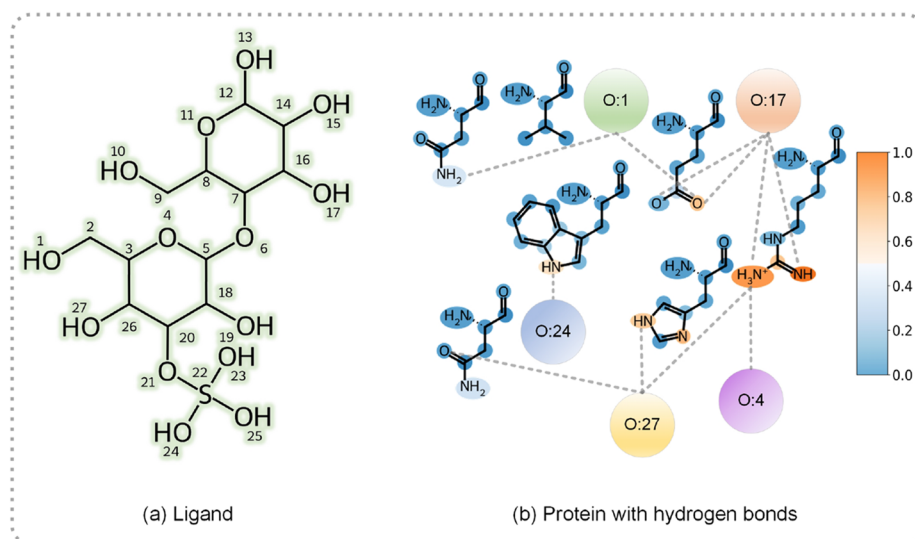


Figure 8. Visualizations of atom importance of the protein using Grad-AAM for the complex with PDB ID 4ym2. (a) The ligand with atom labels. (b) The protein with hydrogen bonds, where O:4 indicates the ligand atom O with atom label 4.

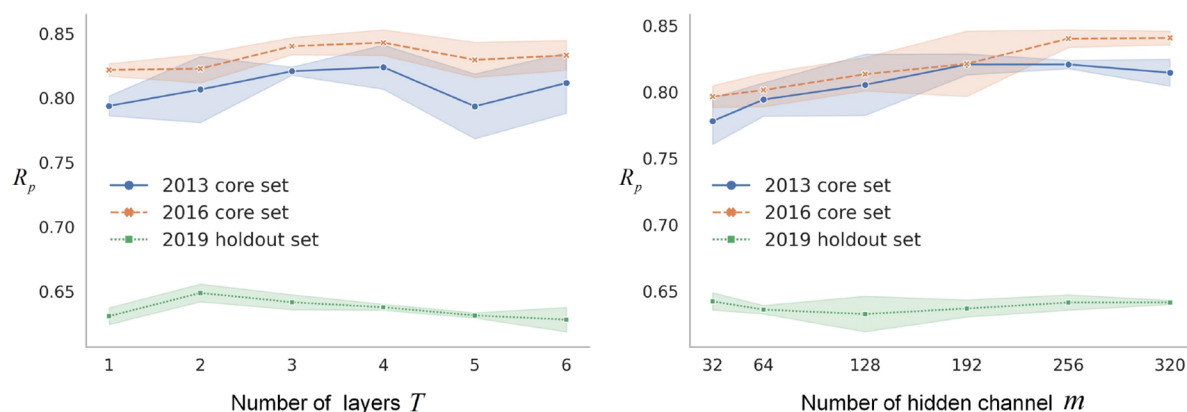


Figure 9. Analysis of parameters T and m on the three external test sets.

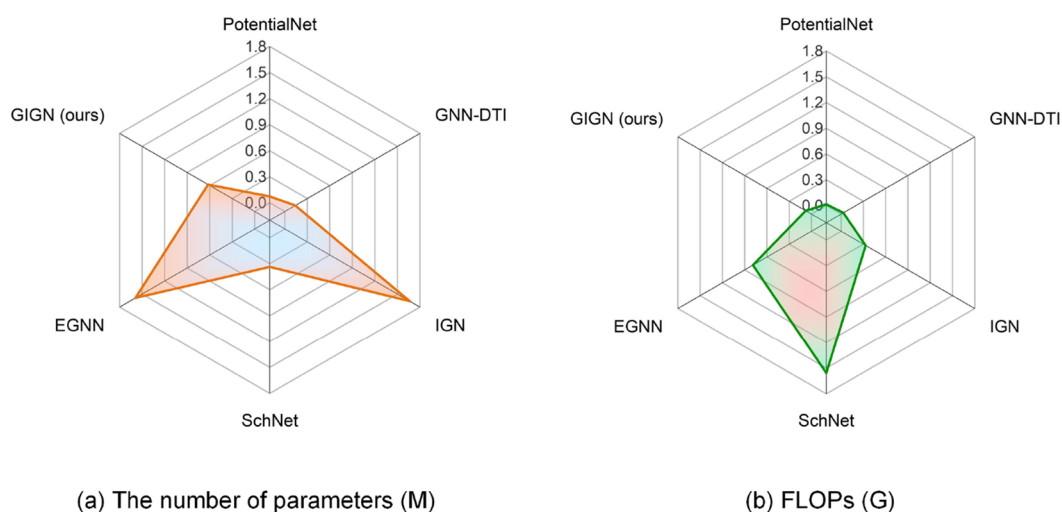


Figure 10. Model complexity in terms of (a) the number of parameters (the lower the better) and (b) FLOPs (the lower the better).

PLA from 3D structures and physical interactions. The core of GIGN is a heterogeneous interaction layer that unifies covalent and noncovalent interactions into the message passing phase while retaining the invariance property with respect to input

translations and rotations, thus avoiding the computational burden of data augmentation. Besides the better generalization capability than other baselines, the visualization results show that GIGN can capture essential features relevant to binding. We

hope that the proposed GIGN can inspire the design of other types of biological 3D interactions.

■ ASSOCIATED CONTENT

Data Availability Statement

Demo, instructions, code, and data for GIGN and other baselines are available at <https://github.com/guaguabujianle/GIGN>.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpclett.2c03906>.

Details of atom features, additional visualization results, and quantitative analysis of the number of covalent and noncovalent interactions (PDF)

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Notes

The authors declare no competing financial interest.

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